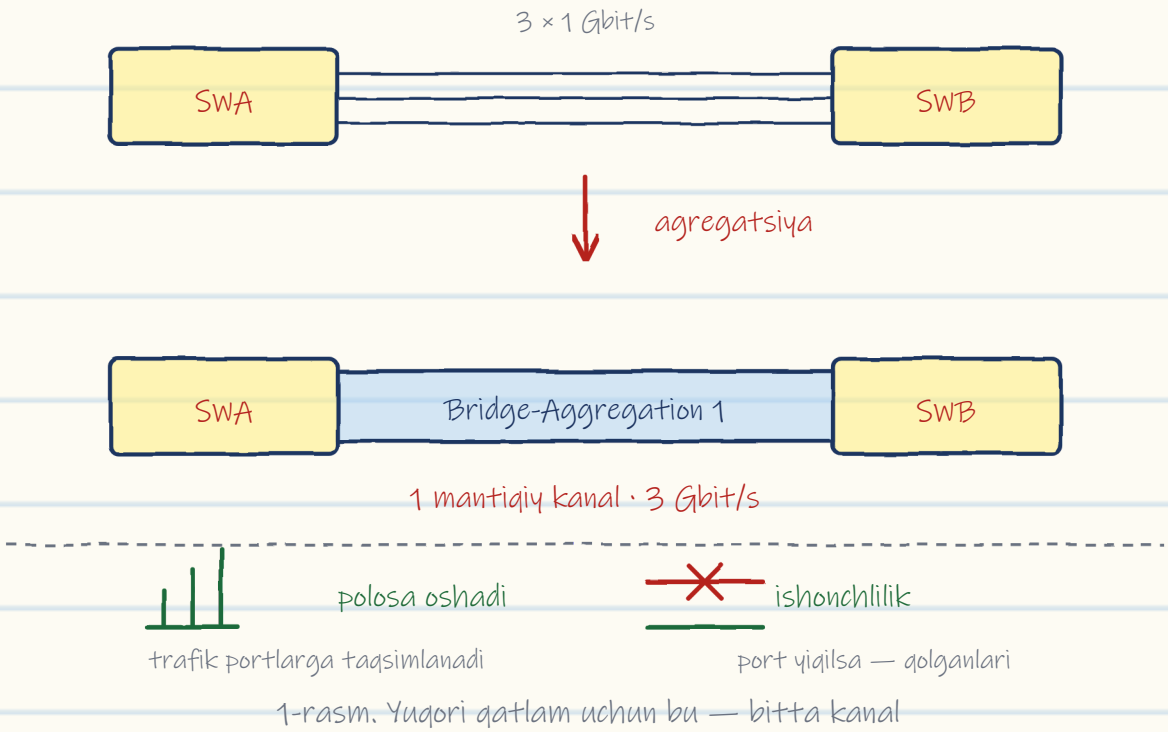
Ethernet kanallarni birlashtirish

Ethernet Link Aggregation

Asosiy atamalar
- Agregatsiya — bir nechta kanalni bitta mantiqiy kanal qilish
- Azo port (member) — guruhga kirgan fizik interfeys
- Yuk taqsimoti — oqimlarni azo portlar orasida bo'lish
- Statik rejim — qo'lda yig'iladi, LACP paket almashmaydi
- LACP — guruhni avtomatik kelishuv protokoli (802.3ad)

1. Agregatsiya nima?

Bir nechta fizik Ethernet porti — bitta mantiqiy kanal (aggregate interface).

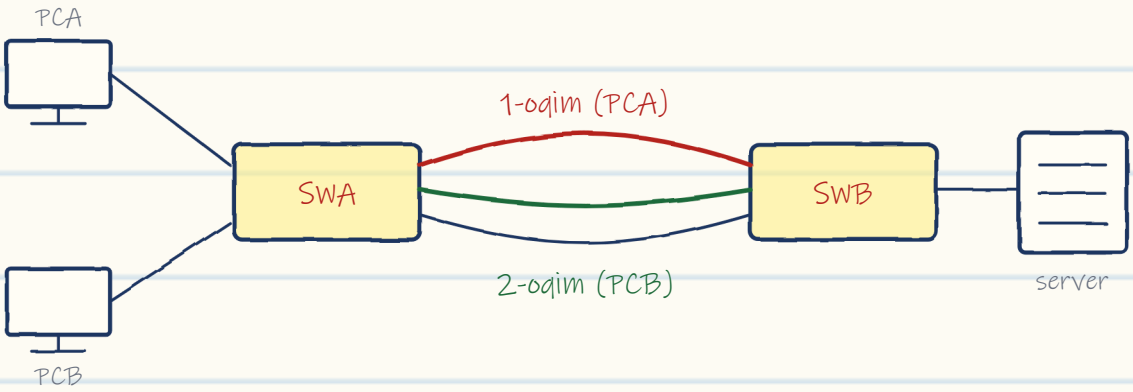


1-rasm. Yuqori qatlam uchun bu — bitta kanal

- a'zo port — guruhga kirgan fizik port
- aggregate interface — yuqori qatlam ko'radigan mantiqiy interfeys

Shart: a'zo portlar bir xil tezlik va rejimda bo'ladi.

2. Yuk taqsimoti



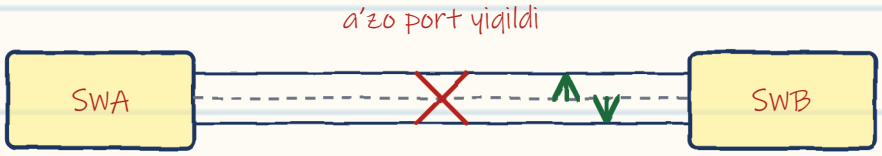
bir oqim — doim bitta port

L2: MAC · L3: IP bo'yicha ajratiladi

2-rasm. Har xil oqim — har xil a'zo port

- oqim — paket maydonlari bo'yicha aniqlanadi
- L2 — manba/qabul MAC; L3 — manba/qabul IP

3. Zaxiralash

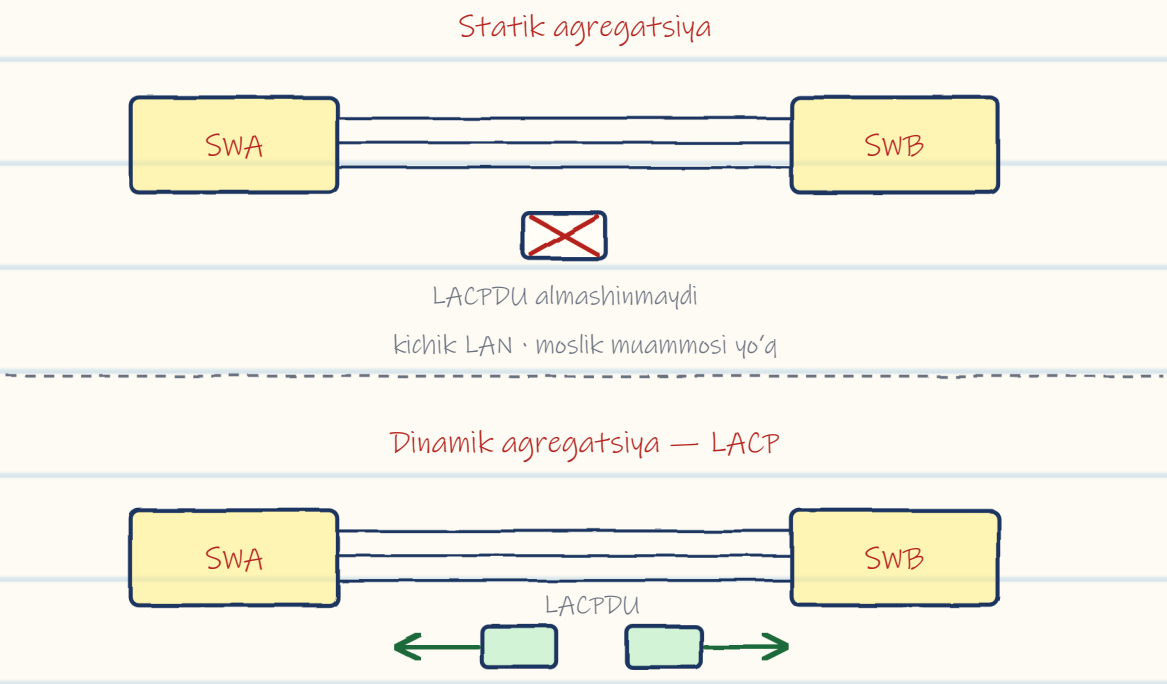


trafik qolgan portlarga o'tadi

yuqori qatlam uchun uzilish sezilmaydi

3-rasm. A'zo portlar bir-birini dinamik zaxiralaydi

4. Ikki usul



4-rasm. Kelishuvsiz va kelishuvli birlashtirish

	Statik	Dinamik
Protokol	yo'q	LACP (802.3ad)
Port holati	kelishilmaydi	kelishiladi
Qachon	kichik LAN	ikki tomon qo'llasa

LACP: ikki tomon LACPDU almashib, qaysi port faol bo'lishini kelishadi.

5. Statik guruh yaratish

- 1) tizim: interface bridge-aggregation raqam — shu raqamli guruh ham tug'iladi
- 2) har bir fizik portda: port link-aggregation group raqam

BAGG 1 — Bridge-Aggregation 1



SWA

```
interface bridge-aggregation 1
interface Ethernet 1/0/1
port link-aggregation group 1
1/0/2 va 1/0/3 uchun ham shunday
```

SWB

```
interface bridge-aggregation 1
interface Ethernet 1/0/1
port link-aggregation group 1
1/0/2 va 1/0/3 uchun ham shunday
```

ikkala tomonda ham bir xil sozlanadi

statik usul kichik LAN'da eng ko'p ishlatiladi

5-rasm. SWA va SWB — uchta port BAGG 1 guruhida

Mantiqiy interfeys	Bridge-Aggregation N (L2)
A'zo port	port link-aggregation group N

Nega statik: protokol paketlari polosani egallamaydi va moslik muammosi chiqmaydi.

6. Holatni ko'rish

display link-aggregation summary

Interface	Mode	Select	Unselect	Share
BAGG1	S	3	0	Shar

S — statik

3 ta selected port

6-rasm. BAGG 1 — statik, 3 ta faol port, yuk taqsimlanadi

- S — statik, D — dinamik; Shar — yuk taqsimlanadi
- faqat selected portlar trafik uzatadi; fizik pastdagi port — unselected

Xulosa

- Agregatsiya: bir nechta fizik port → bitta mantiqiy kanal
- Foyda: polosa oshadi + a'zo portlar bir-birini zaxiralaydi
- Yuk taqsimoti oqim bo'yicha: L2 — MAC, L3 — IP
- Statik — kelishuvsiz; dinamik — LACP, IEEE 802.3ad
- display link-aggregation summary — holatni ko'rsatadi

Tekshirish

1. Agregatsiyaning 2 foydasi? — polosa va ishonchlilik
2. LACP standarti? — IEEE 802.3ad
3. Oqim nima bo'yicha aniqlanadi? — MAC (L2) / IP (L3)
4. Qaysi portlar trafik uzatadi? — faqat selected